Ethan Hosier

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Education

Imperial College London — London, UK

MEng Computing - First Class Honours — *Oct. 2022 – June 2026*

Experience

Palantir | *Python, Spark, TypeScript, React.js* — April 2025 – September 2025

Forward Deployed Software Engineer, Intern — *London, UK*

- Rewrote a legacy data pipeline processing billions of customer transactions every hour, achieving 3.5× faster runtime through caching and optimized parallel execution across distributed Spark clusters
- Transformed the client's financial forecasting process by developing a spreadsheet engine from scratch supporting real-time multi-user collaboration, formulas, keyboard shortcuts and configurable templates
- Built an AI model router for Palantir's internal chat system with 4,000+ users, automatically selecting the optimal language model for each conversation
- Led meetings with the client's executives and lead engineers, and iterated on features directly with end users

Revolut | *Java, TypeScript, React.js, Kafka, Google Cloud Platform* — July 2024 – September 2024

Software Engineer, Intern — *London, UK*

- Developed the backend of an Authorised Data Source consolidating product and pricing-rule configurations into a single interface—used daily by the CEO and leadership team—eliminating manual database queries
- Reduced query latency by 99% and cut DB reads to near-zero by consuming product/pricing change events from a Kafka topic into a local in-process cache and serving price calculations directly from the cache instead of the DB.
- Integrated with internal services including the Legal Service (document assignment to tax configurations) and the People Platform (employee data in the admin app), ensuring seamless interaction across core business systems

Projects

PintOS Operating System Kernel | *C*

- Implemented demand paging with clock-based page replacement, swapping, stack growth and memory-mapped files
- Added argument passing and process creation/wait handling for user programs, and implemented system calls for process control and I/O
- Implemented priority-based kernel thread scheduling with priority donation and a multilevel feedback queue

WACC Compiler | *Scala*

- Implemented WACC parsing and semantic checks in Scala, including type checking and scoped symbol resolution
- Generated x86-64 assembly for functions, arrays, pairs, and control flow, with stack and register allocation
- Implemented constant folding, function inlining, dead code elimination and peephole optimisations, reducing execution time by 12%

Technical Skills

Java, Go, Python, TypeScript, Docker, AWS, React.js, Next.js, Redis, Terraform, Spark, Kafka